# Defining the thermal comfort range of UTCI, PET, and mPET thermal indices

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Deutscher Wetterdienst
Wetter und Klima aus einer Hand

DWD

# Introduction

- Research Purpose:

  Discussion about the thermal comfort range

- Process:

  Use the database of 2,135 questionnaires from Taiwan,

  and establish a nine-level (cold to hot) thermal stress classification into the past 10-year climate data.

- Measurement time:

  The thermal comfort database is based on the Lin (2015) study.

  Long-term climate data since 2001 to 2011, Tainan weather station.

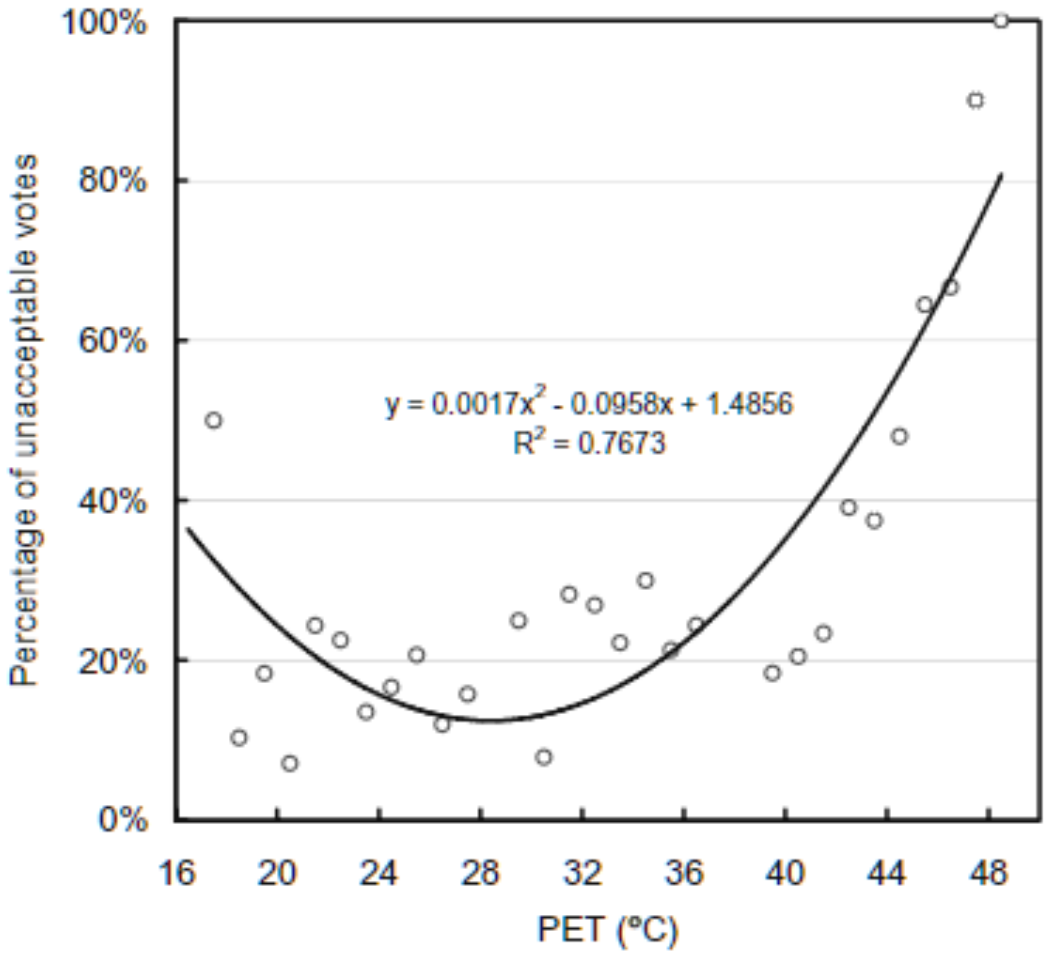



**Fig. 5** Thermal comfort range for outdoor environments in Taiwan

**Table 2** Thermal sensations and PET classes for Taiwan and Western/Middle European classes

| Thermal sensation | PET range for Taiwan (°C PET) | PET range for Western/Middle European[a] (°C PET) |
|---|---|---|
| Very cold | | |
| | 14 | 4 |
| Cold | | |
| | 18 | 8 |
| Cool | | |
| | 22 | 13 |
| Slightly cool | | |
| | 26 | 18 |
| Neutral | | |
| | 30 | 23 |
| Slightly warm | | |
| | 34 | 29 |
| Warm | | |
| | 38 | 35 |
| Hot | | |
| | 42 | 41 |
| Very hot | | |

[a] Source: Matzarakis and Mayer (1996)

Tourism climate and thermal comfort in Sun Moon Lake, Taiwan (2008)

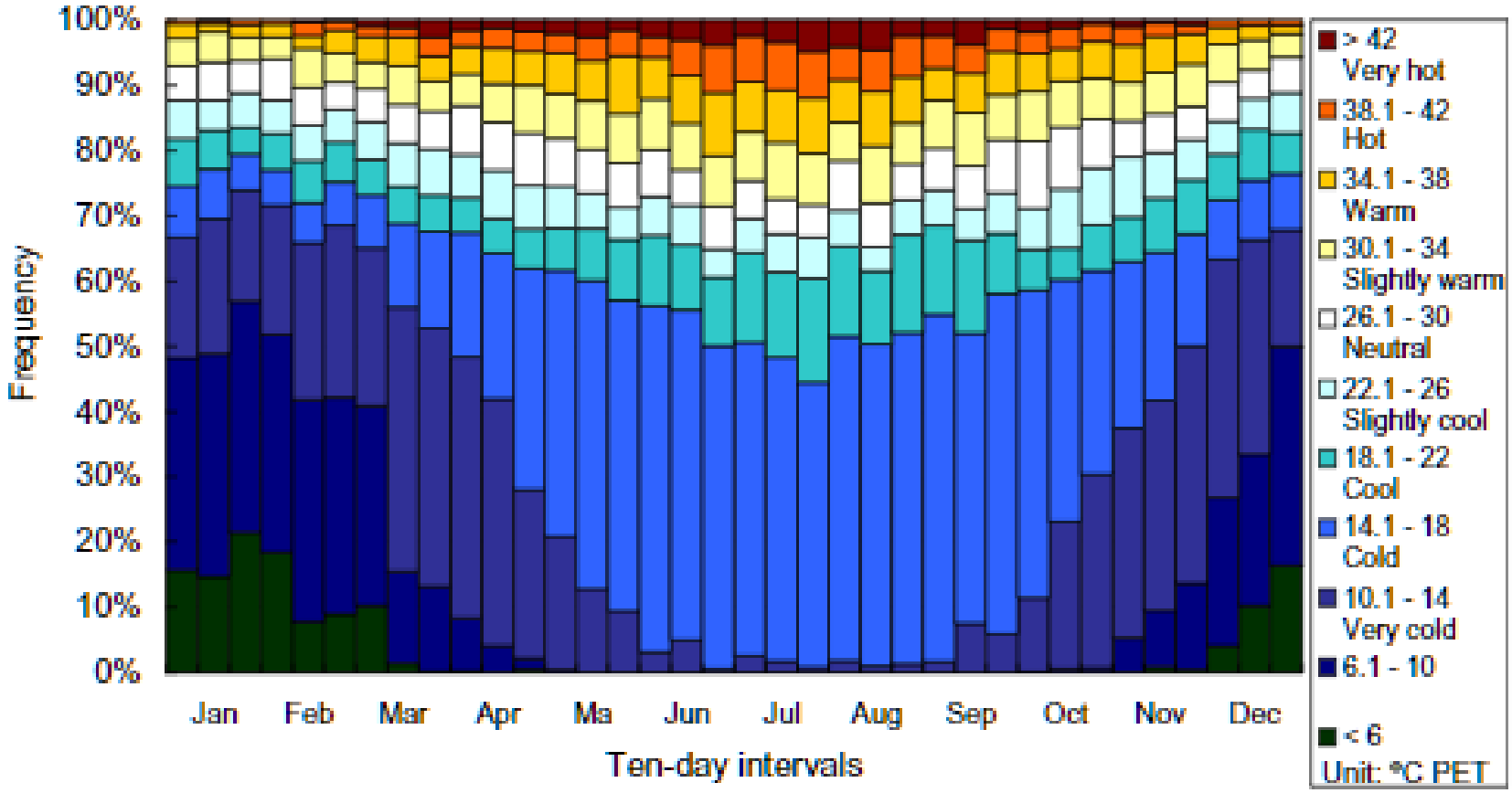



Fig. 7 PET frequencies with Taiwan Scale for 1996–2005. The comfort period (“slightly cool”to“slightly warm”) for Taiwan residents is onlyachieved during 20% of the year

Tourism climate and thermal comfort in Sun Moon Lake, Taiwan.

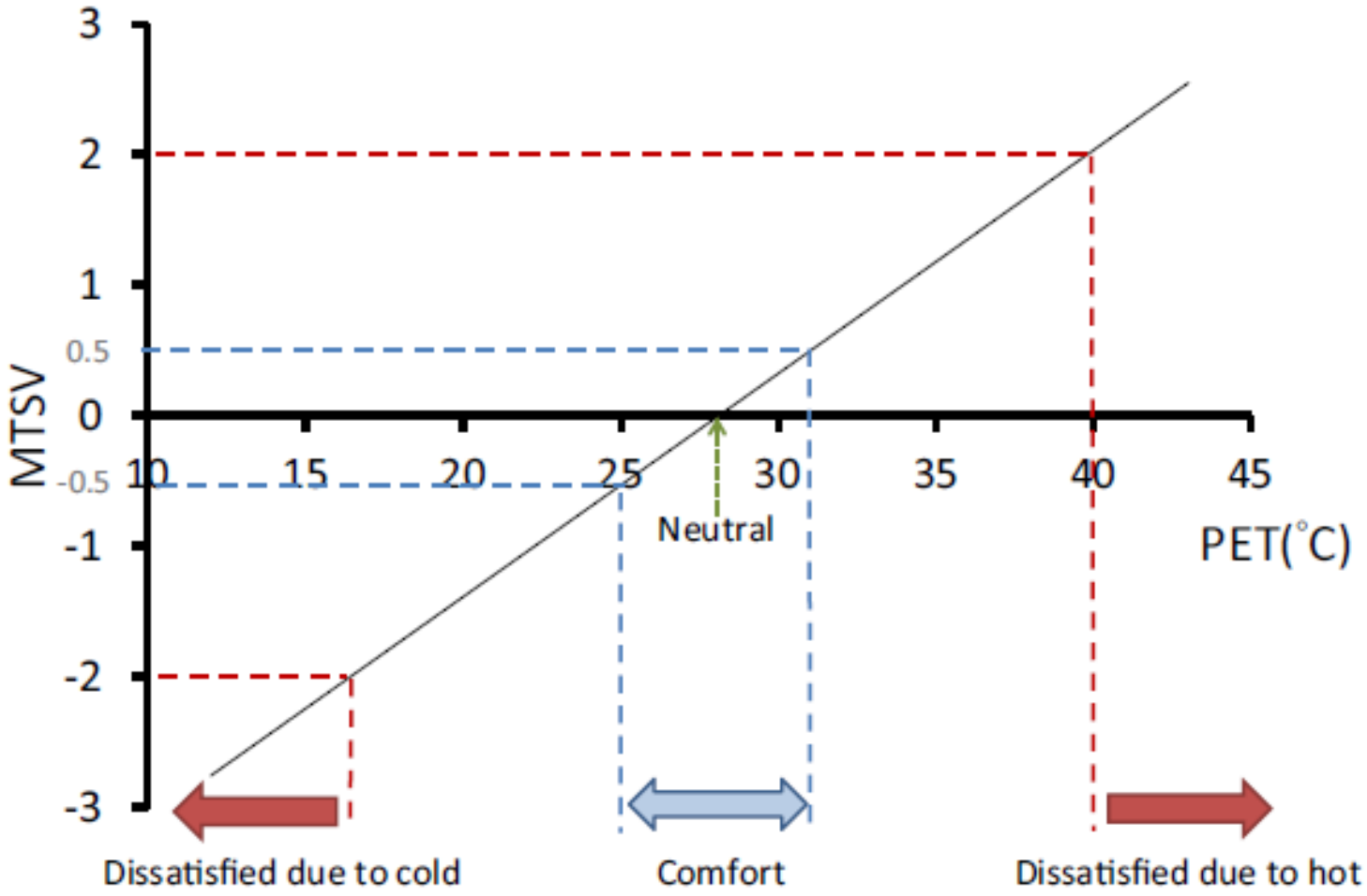



**Fig. 3** Schematic diagram of the method for obtaining the neutral value, comfort, and dissatisfied ranges

Customized rating assessment of climate suitability (CRACS):
climate satisfaction evaluation based on subjective perception (2015)

**Table 1** Neutral value, comfort range, and dissatisfied arranges for each climate parameter

| | Neutral value | | Comfort range | | Dissatisfied arrange | |
|---|---|---|---|---|---|---|
| | Perception value[a] | Neutral value | Perception range | Parameters range | Perception range | Parameters range |
| PET [°C] | MTSV=0 | 28.1 | MTSV=−0.5–0.5 | 25–31 | MTSV>2 | PET>40 |
| | | | | | MTSV<−2 | PET<16 |
| Air temperature [°C] | MASV=0 | 26.2 | MASV=−0.5–0.5 | 24–28.4 | MASV>2 | Ta>35 |
| | | | | | MASV<−2 | Ta<17.3 |
| Relative humidity [%] | MHSV=0 | 63.6 | MHSV=−0.5–0.5 | 54–73 | MHSV>1 | RH>82.7 |
| | | | | | MHSV<−1 | RH<44.4 |
| Wind speed [m/s] | MWSV=0 | 2.9 | MWSV=−0.2–0.2 | 2.2–3.5 | MWSV>0.5 | $v$>4.4 |
| | | | | | MWSV<−0.5 | $v$<1.3 |
| Global radiation [W/m$^2$] | MSSV=0 | 454.8 | MSSV=−0.7–0.7 | 200–700 | MSSV>0.7 | $G$>700 |
| | | | | | MSSV<−1 | $G$<98 |
| Cloud cover [–] | MCSV=0 | 6.7 | MCSV=−0.50–0 | 3–6 | MCSV>0.1 | Cloud>7 |
| | | | | | MCSV<−0.6 | Cloud<2 |
| Precipitation [mm] | MRSV=0 | 2.4 | MRSV=−0.8–0 | 0–0.05 | MRSV>1 | Pr>0.15 |

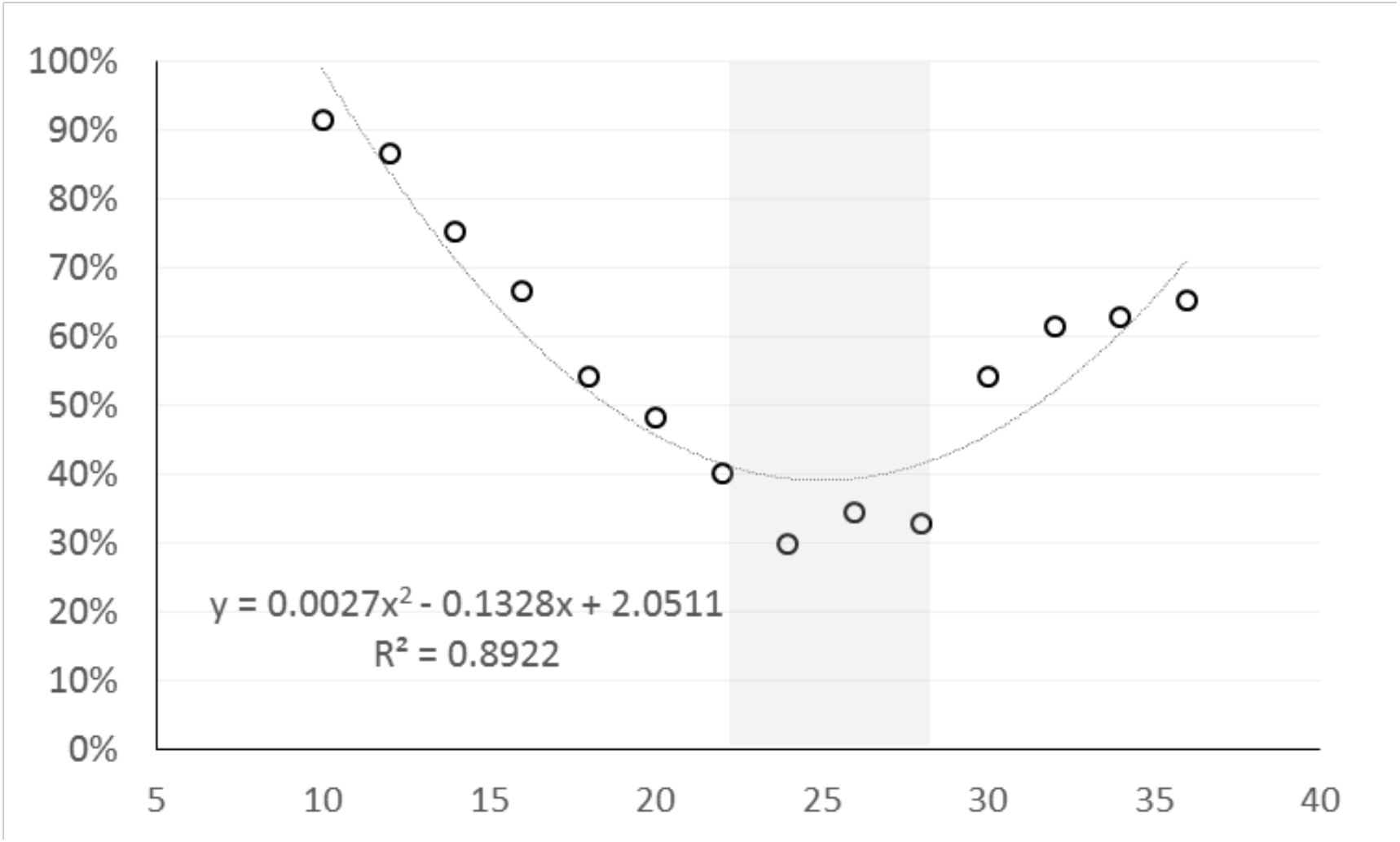
[a] Mean sensation vote for overall thermal environment (MTSV), air temperature (MASV), relative humidity (MHSV), wind speed (MWSV), solar radiation (MSSV), cloud cover (MCSV), and amount of precipitation (MRSV)

Customized rating assessment of climate suitability (CRACS):
climate satisfaction evaluation based on subjective perception (2015)

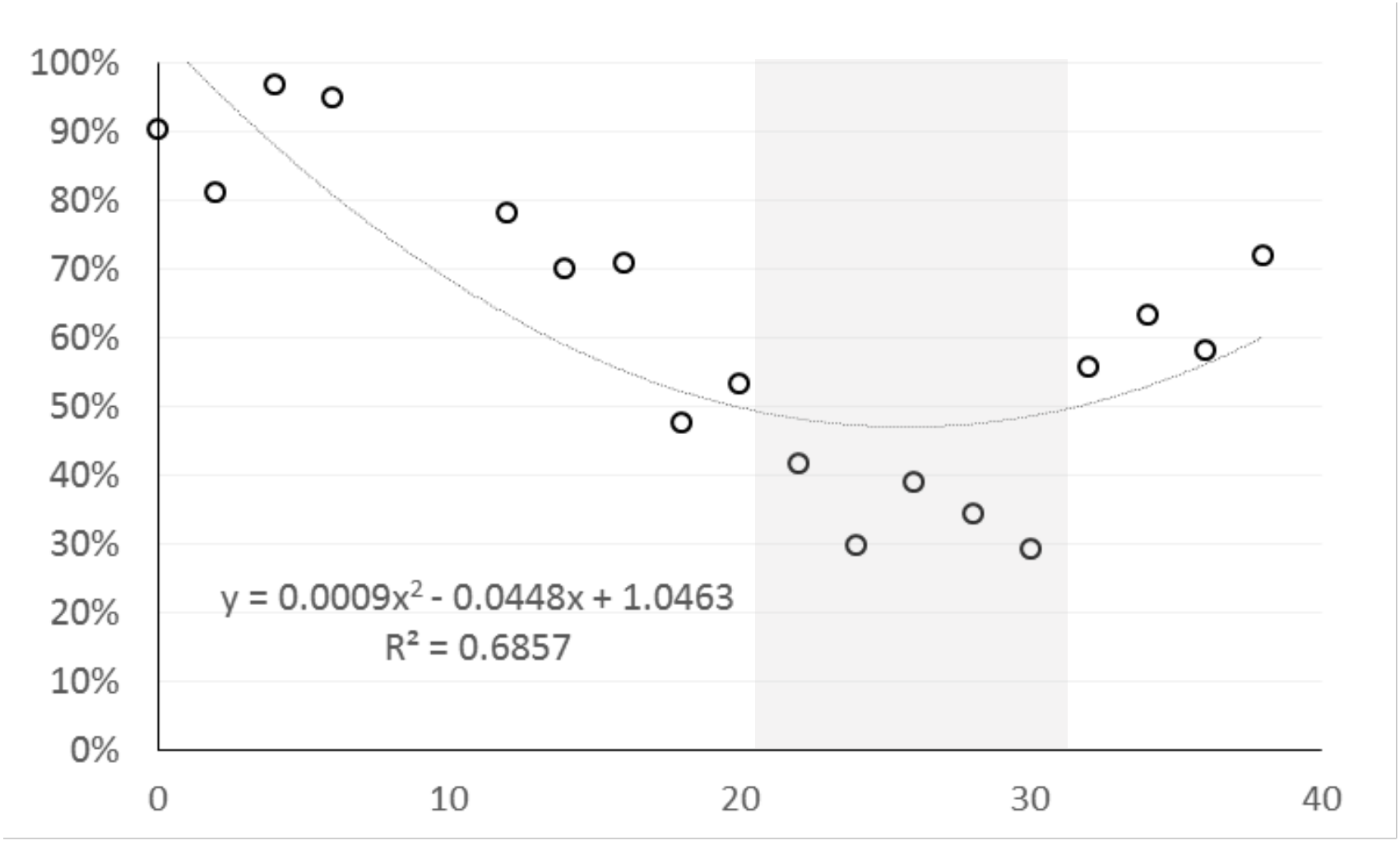
# Method_ index data

| | Temperature | Wind speed | Black globe temperature | Relative Humidity | Range |
|---|---|---|---|---|---|
| PET/mPET | Measurement Data | Weather station *0.588 | Measurement Data | Measurement Data | Question naire |
| UTCI | Measurement Data | Weather station | Measurement Data | Measurement Data | Question naire |

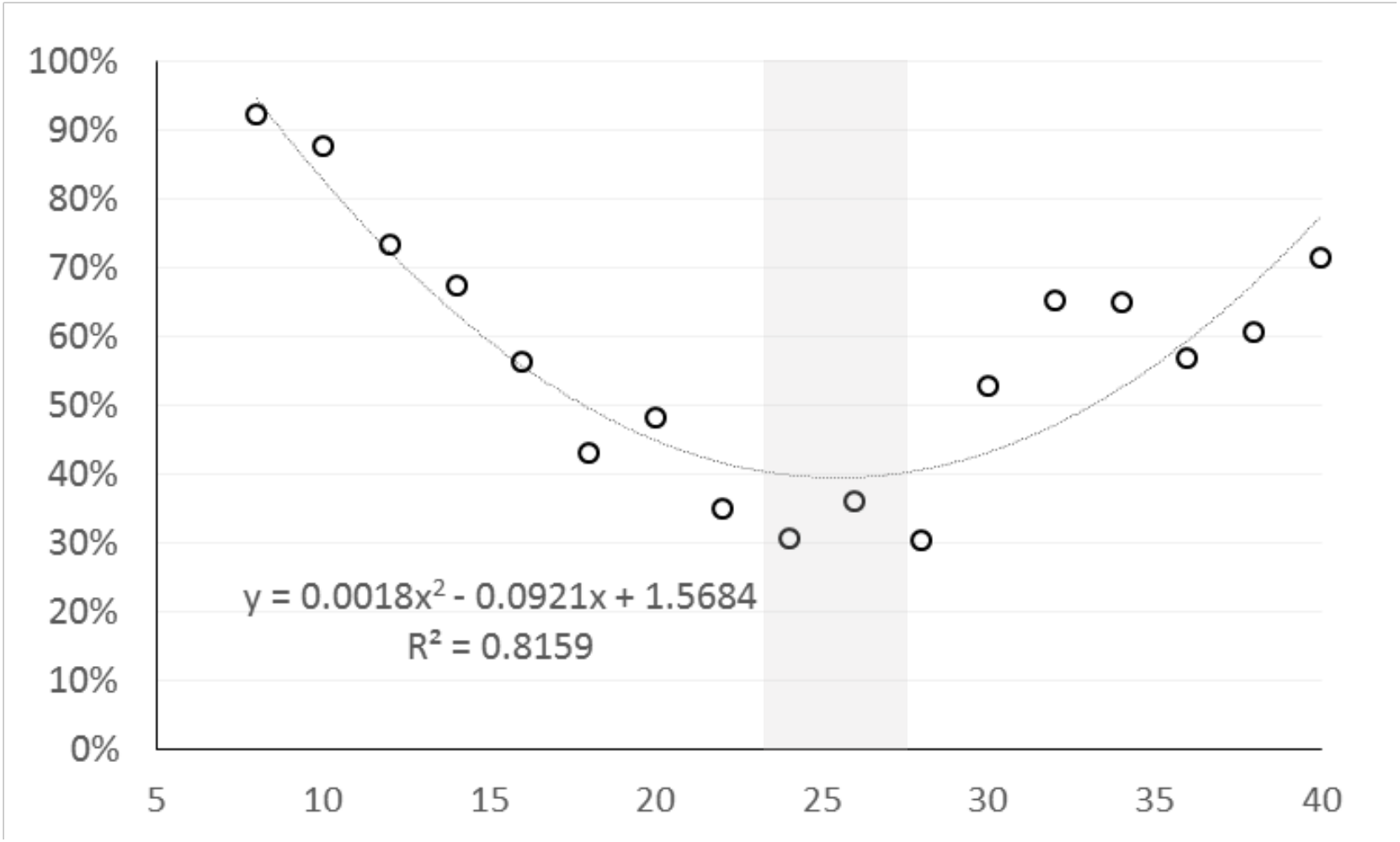
# mPET

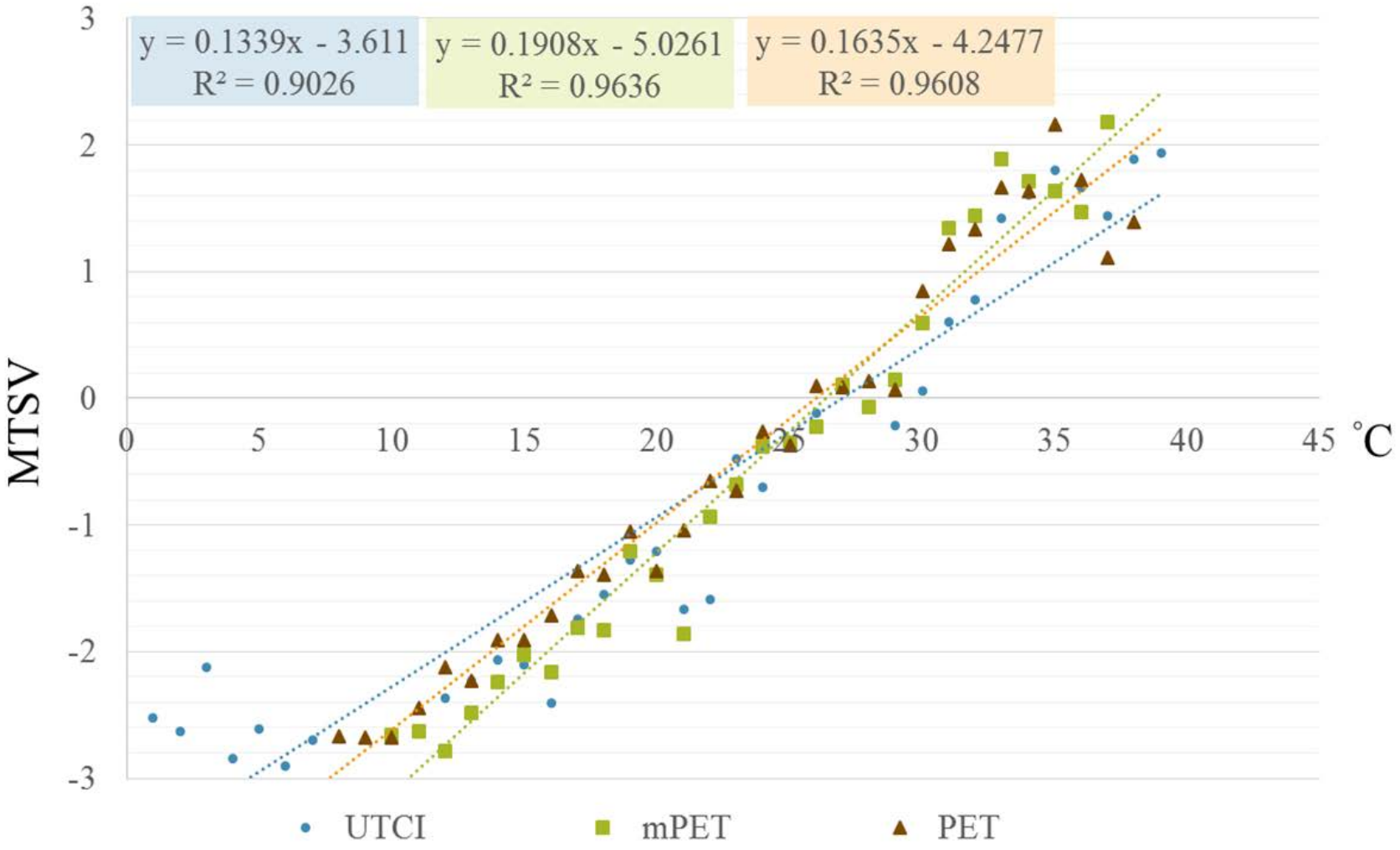

100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
5
10
15
20
25
30
35
40
y = 0.0027x² - 0.1328x + 2.0511
R² = 0.8922

# UTCI


100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
0
10
20
30
40
y = 0.0009x² - 0.0448x + 1.0463
R² = 0.6857

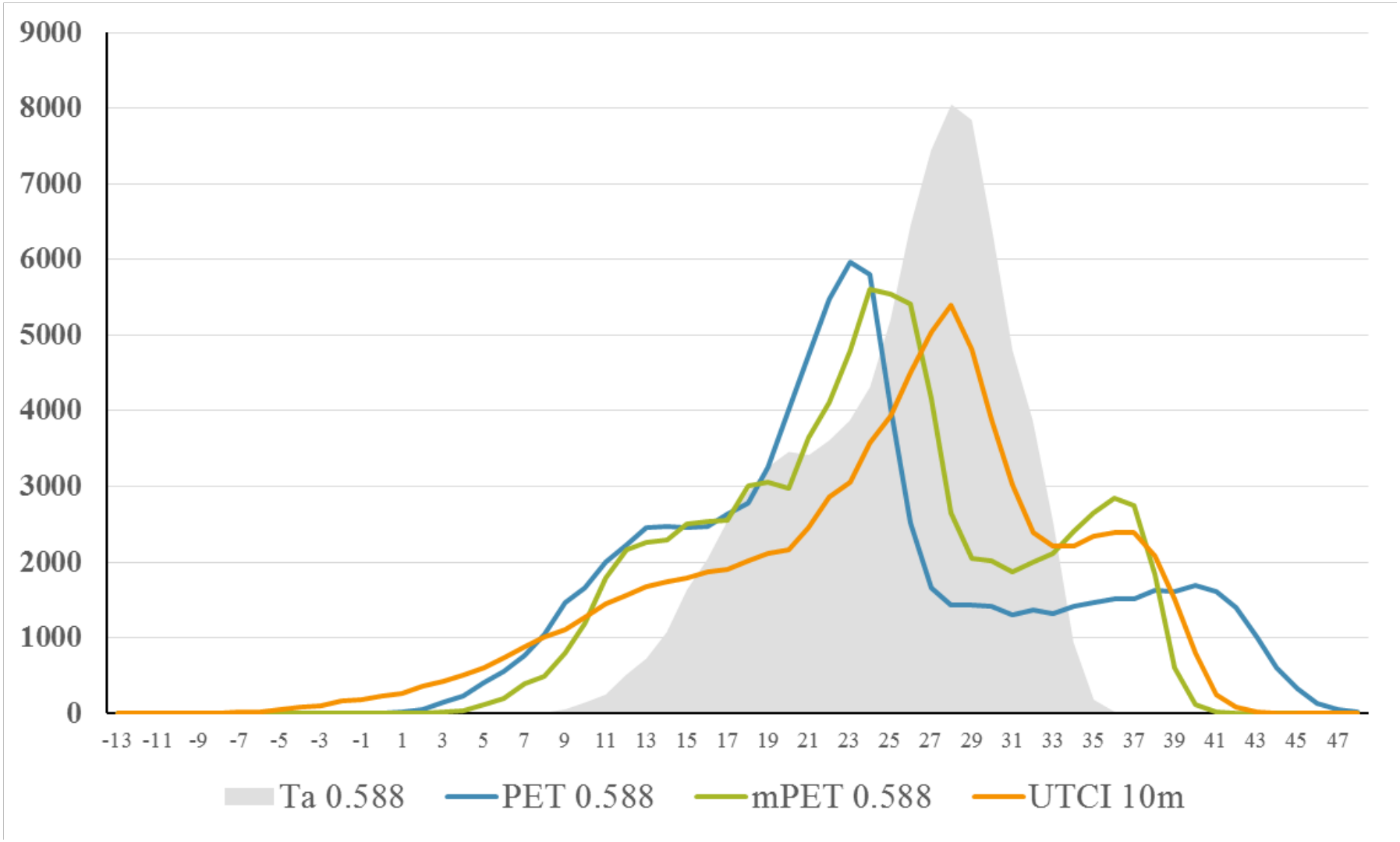
# PET


100%
90%
80%
70%
60%
50%
40%
30%
20%
10%
0%
5
10
15
20
25
30
35
40
y = 0.0018x² - 0.0921x + 1.5684
R² = 0.8159

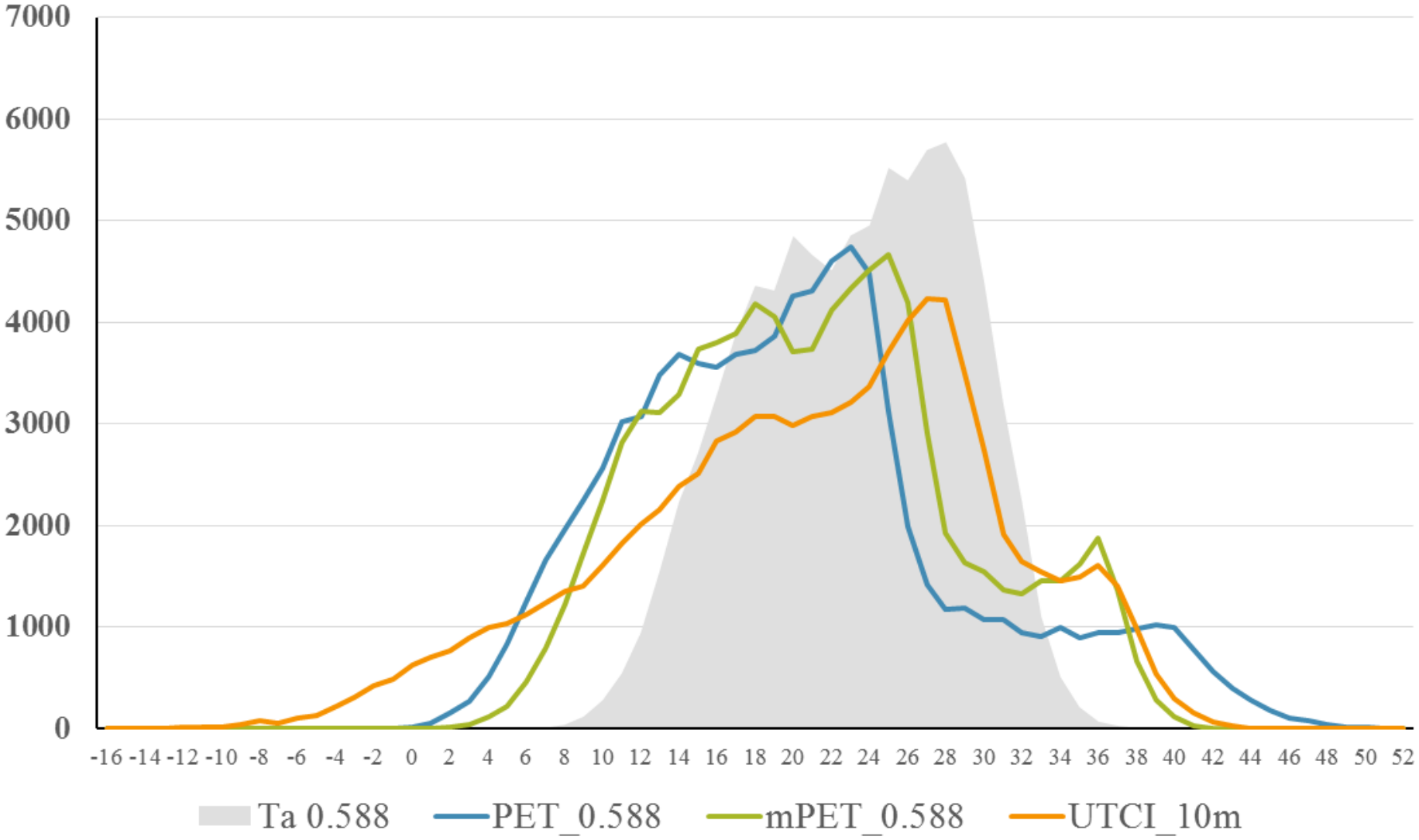

y = 0.1339x - 3.611
R² = 0.9026
y = 0.1908x - 5.0261
R² = 0.9636
y = 0.1635x - 4.2477
R² = 0.9608
MTSV
3
2
1
0
-1
-2
-3
0
5
10
15
20
25
30
35
40
45
°C
UTCI
mPET
PET

| | MTSV | mPET (Taiwan,2015) | PET (Taiwan,2015) | PET (Europe) | PET (Taiwan,2008) | UTCI (Taiwan,2015) | Heat stress | UTCI (Europe) |
|---|---|---|---|---|---|---|---|---|
| Very cold | <-3 | <8 | <5 | <4 | <14 | <1 | Extreme cold stress | <-40 |
| Cold | -3 | 8-13 | 5-11 | 4-8 | 14-18 | 1-8 | Very strong cold stress | -27 to -40 |
| Cool | -2 | 13-18 | 11-17 | 8-13 | 18-22 | 8-16 | Strong cold stress | -27 to -13 |
| Slightly cool | -1 | 18-24 | 17-23 | 13-18 | 22-26 | 16-23 | Moderate cold stress | -13 to 0 |
| Comfortable | 0 | 24-29 | 23-29 | 18-23 | 26-30 | 23-31 | Slight cold stress | 0 to 9 |
| Slightly warm | 1 | 29-34 | 29-35 | 23-29 | 30-34 | 31-38 | No thermal stress | 9 to 26 |
| Warm | 2 | 34-39 | 35-41 | 29-35 | 34-38 | 38-46 | Slight heat stress | 26 to 32 |
| Hot | 3 | 39-45 | 41-47 | 35-41 | 38-42 | 46-53 | Moderate heat stress | 32 to 38 |
| Very hot | >3 | >45 | >47 | >41 | >42 | >53 | Strong heat stress | 38 to 40 |
| | PET (Europe,1996) ：(Matzarakis and Mayer 1996)<br>PET (Taiwan,2008)：(Lin and Matzarakis 2008)<br>UTCI (Europe) ：(Błażejczyk et al. 2013)<br>mPET/PET/UTCI (Taiwan,2015) ： (Lin et al. 2015) | | | | | | Very strong heat stress | 40 to 46 |
| | | | | | | | Extreme heat stress | >46 |

# Thermal range number distribution in Tainan



# Thermal range number distribution in Keelung



# Discussion

- Further investigation of the thermal comfort ranges in Taiwan and Europe showed that for the PET, the lowest acceptable (-2) temperature in Taiwan was 10°C higher than that in Europe, indicating that residents of subtropical regions are less tolerant of low temperatures.

- For the UTCI, the thermal comfort ranges recommended worldwide differed considerably from the Taiwanese ranges derived from local questionnaire data. This result implies that the thermal comfort range of thermal comfort indices should be based on the results obtained from local questionnaires to meet the local's needs.

# Thank you for your attention

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